Maxim Qu

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Education

McGill University Montreal, CA
B.Sc. Software Engineering *Aug. 2024 – Apr. 2027*
- Relevant Coursework: Machine Learning, Data Structures and Algorithms, Database Systems, Software Design

Technical Skills

Programming Languages: Python, Java, C/C++, Dart, OCaml, Assembly, Bash, JavaScript, HTML/CSS, SQL
Hardware & Systems: PC Building, Hardware Diagnostics, Component Isolation, Stress Testing, Benchmarking, Thermal Monitoring, GPU/CPU Tuning, Windows
Frameworks & Libraries: Flutter, React, Vite, Node.js, Flask, SQLAlchemy, Tailwind CSS, Drift
Developer Tools & Cloud: Git, GitHub, Supabase, PostgreSQL, Terraform, VS Code, IntelliJ, Figma
Spoken Languages: English (Fluent), French (Fluent), Mandarin (Fluent)

Experience

Software Developer Intern Montreal, CA
Mada Community Center *May 2026 – July 2026*
- Developed a mobile app in **Flutter**, backed by an 18-table **Supabase** schema with Row-Level Security (RLS).
- Built offline data persistence with **Drift** and Supabase Edge Functions, enabling the app to maintain functionality across intermittent connectivity.
- Implemented user profile switching, reading activity tracking, and an in-app reward wallet based on **Figma** designs.
- Structured the codebase around modular components to simplify maintenance and future feature development

Projects

Turing Blend — Devpost | *Interactive 3D Web App using React, Three.js, Gemini API* June 2026
- Won **Best Use of Gemini API** at MPC Hacks.
- Engineered 5 AI opponents with **Gemini 2.0 Flash** with distinct accusation, defense, and voting behaviors.
- Built an interactive 3D interrogation room in **Three.js** featuring a roundtable, dynamic lighting, and animated character sprites.
- Secured **ElevenLabs** API integration through a Vite server proxy, keeping API credentials out of the client

Mixxx DJ Software | ***Open Source*** *Contribution using C++, Qt, CMake, Visual Studio, Git* March 2026
- Diagnosed a **Qt** framework assertion failure in debug builds triggered by an unhandled empty list.
- Submitted a GitHub PR implementing the fix which was approved, and merged into Mixxx's upstream main branch.

Moris Discord URL Scanner Bot | *Discord Bot using Python, Discord API, Regex* Aug. 2025
- Won **Cybersecurity Bootcamp Project** competition with CCubed for automated threat detection tooling.
- Implemented automated message listener in Python using regex to detect URLs in Discord channels in real time.
- Integrated the **VirusTotal API** to submit extracted URLs, poll scan results across 70+ security vendors, and quarantine known IP loggers with dynamic status alerts.

Patiently — Devpost | *Web app using Python, Flask, SQLAlchemy, Terraform* Jan. 2025
- Won **Best Use of Terraform** at McHacks 12.
- Implemented Flask REST endpoints, **OAuth** authentication, and relational database models with SQLAlchemy.
- Built a predictive occupancy and wait-time visualization system for hospital emergency rooms.

Custom PC Engineering & Hardware Optimization | *Hardware Diagnostic, Benchmarking* 2021 – Present
- Built and diagnosed **8+** custom PCs, using component isolation, stress testing, benchmarking, and thermal monitoring to troubleshoot hardware issues
- Reduced a RX 6600xt GPU temperatures from 80°C to low 60s using custom cooling system with **3D-printed** fan brackets and dual 120 mm Noctua fans, substantially reducing system noise.
- Set a GPU **undervolt** that reduced power consumption from 140W to 90W without affecting gaming performance.

Others

Interest: Music Production, Classical Piano, John Coltrane
Experience: 3 years of Piano teaching experience
Leadership: 40 hours of Volunteering at OCCSC